Digital Forensics & Incident Response (DFIR)

1. NTFS MFT Forensic Parser & Deleted File Analysis Tool

Overview: Developed a digital forensic tool to parse and analyze NTFS Master File Table (MFT) records for identifying deleted and suspicious files from Windows disk images. The tool enables investigators to reconstruct file system activity and detect potential evidence removal attempts.

Approach / Implementation

• Built an automation wrapper around MFTECmd to extract detailed NTFS metadata
• Implemented logic to identify deleted file records using IsInUse flag analysis
• Parsed key forensic attributes including file paths, timestamps (MACB), file size, and record metadata
• Structured extracted data for efficient filtering and investigation
• Designed workflow to support forensic timeline reconstruction

Technologies Used: C++, MFTECmd, Windows File System Internals, JSON/Data Structuring

Key Features

• Automated MFT parsing and metadata extraction
• Detection of deleted file entries
• Extraction of MACB timestamps for timeline analysis
• Structured reporting for forensic investigations

Outcome / Impact: The tool significantly improves the efficiency of forensic investigations by automating the identification of deleted files and enabling analysts to reconstruct file system activity. It supports incident response workflows by helping detect potential data tampering and evidence removal attempts.

2. RAM Dump Analysis Framework

Overview: Developed a memory forensics analysis framework for investigating RAM dump images to identify suspicious processes, injected code, and hidden artifacts. The framework automates the extraction and correlation of memory-based forensic evidence to support incident response investigations.

Approach / Implementation

• Integrated Volatility3 plugins to automate memory artifact extraction
• Developed scripts to analyze process lists, network connections, loaded DLLs, and registry data
• Implemented correlation logic to link multiple artifacts for deeper analysis
• Automated workflows to reduce manual forensic effort
• Designed the framework to support structured and repeatable investigations

Technologies Used: Python, C++, Volatility3, Memory Forensics Techniques, Winpmem

Key Features

- Automated RAM dump analysis
- Detection of suspicious and hidden processes
- Extraction of network and process artifacts
- Correlation of memory-based forensic data
- Support for incident response workflows

Outcome / Impact: The framework improves the efficiency of memory forensics investigations by automating artifact extraction and analysis. It enables faster identification of malicious processes and persistence mechanisms, supporting effective incident response and threat detection.

3. Live Memory Forensics Tool

Overview: Developed a live memory forensics tool designed to collect and analyze runtime system information during active incident response. The tool enables investigators to inspect running processes, loaded modules, and memory regions to detect in-memory and fileless threats.

Approach / Implementation

- Implemented modules to analyze active processes and loaded system components
- Integrated memory acquisition utilities to capture RAM data during runtime
- Developed logic to identify suspicious memory regions and anomalies
- Enabled inspection of live system state without requiring full system shutdown
- Designed the tool for use during active incident response scenarios

Technologies Used: C++, Windows Internals, Memory Acquisition Techniques, nlohmann/json, WMI,Win32API

Key Features

- Live system memory inspection
- Detection of fileless and in-memory malware
- Analysis of running processes and modules
- Runtime memory acquisition support
- Designed for real-time incident response

Outcome / Impact: The tool enhances incident response capabilities by enabling investigators to analyze volatile memory in real time. It supports the detection of advanced threats such as fileless malware that do not leave traces on disk, improvingoverall forensic visibility.

4. Windows Prefetch Forensic Parser & Execution Analysis Tool

Overview: Developed a forensic analysis tool to parse and analyze Windows Prefetch artifacts for investigating application execution history. The tool enables extraction of execution patterns and supports detection of suspicious or unauthorized program activity.

Approach / Implementation

- Built a Python wrapper around PECmd to automate Prefetch parsing

• Extracted execution-related artifacts including application paths, run counts, and timestamps
• Processed referenced file dependencies associated with execution
• Structured extracted data using JSON for efficient analysis
• Generated human-readable HTML reports for investigation workflows

Technologies Used: Python, PECmd, JSON (nlohmann/json), Windows Forensic Artifacts

Key Features
• Automated parsing of Prefetch files
• Extraction of execution history and run frequency
• Identification of application execution patterns
• JSON-based structured forensic output
• HTML report generation for easy analysis

Outcome / Impact: The tool improves forensic analysis by automating the interpretation of Prefetch artifacts, enabling investigators to quickly identify suspicious execution behavior and reconstruct user activity timelines during incident response.

5. Windows Forensic Artifact Collector

Overview: Developed a forensic triage tool to automate the collection of critical system artifacts from Windows machines during incident response. The tool enables rapid acquisition of relevant forensic data to support early-stage investigation and threat detection.

Approach / Implementation
• Developed modules to collect system-level artifacts including processes, network configuration, logs, and registry data
• Automated data collection workflows to reduce manual effort
• Structured collected data for efficient analysis
• Designed the tool to support rapid triage during active investigations
• Focused on minimizing system impact during data collection

Technologies Used: Python, Windows APIs, System Forensics Techniques C++

Key Features
• Unified forensic investigation platform
• Support for multiple evidence formats (RAM, DD, E01)
• Integrated memory and disk analysis
• Structured investigation workflows
• Cross-artifact correlation for deeper insights

Outcome / Impact: The tool accelerates the initial phase of incident response by enabling fast and reliable collection of forensic evidence. It helps investigators quickly assess compromised systems and prioritize further analysis.

6. RapiDFIR – AI-Driven Digital Forensics & Incident Response Platform

Overview: Developed a comprehensive digital forensics platform designed for conducting end-to-end investigations on Windows systems. The platform integrates memory analysis, disk forensics, and evidence processing to support efficient incident response and forensic workflows.

Approach / Implementation

- Designed a centralized forensic platform for remote evidence collection and analysis
- Integrated workflows for memory forensics, disk analysis, and artifact processing
- Enabled support for RAM dumps, raw disk images (dd), and E01 evidence files
- Implemented structured case management for organizing investigations
- Incorporated AI-driven analysis concepts to assist in identifying suspicious patterns
- Developed mechanisms to reduce dependency on on-site forensic teams

Technologies Used: Python, C++, Digital Forensics Tools, AI-based Analysis Concepts, Evidence Processing Techniques

Key Features

- Remote forensic data collection
- AI-assisted investigation workflows
- Centralized case management system
- Support for multiple evidence formats (RAM, DD, E01)
- Real-time analysis capabilities
- Integrated memory and disk forensics

Outcome / Impact: Rapid FIR enhances incident response efficiency by enabling rapid, remote, and centralized forensic investigations. It reduces operational overhead, accelerates evidence analysis, and improves the ability to detect and respond to cyber threats in real time.

Network Security & Monitoring

Projects focused on monitoring network activity, analyzing system-level communication, and correlating processes with network behavior to detect suspicious or malicious activity.

7. Network Tracer & Process Correlation Engine

Overview: Developed a real-time network monitoring and tracing system that maps active network connections to their originating processes. The tool enables analysts to identify suspicious communication patterns and detect potential command-and-control (C2) activity.

Approach / Implementation

- Implemented packet interception using WinDivert to capture real-time network traffic
- Developed correlation logic linking network activity with process identifiers (PID, PPID)
- Extracted executable paths, connection duration, and traffic statistics
- Captured detailed telemetry including source/destination IP and data transfer metrics
- Designed system for continuous monitoring and analysis

Technologies Used: C++, Win Divert, Win32 API, Network Monitoring Techniques, Python(QT5)

Key Features

- Real-time network traffic interception
- Process-to-network correlation
- Detection of suspicious outbound connections
- Detailed traffic and connection analysis
- Identification of potential C2 communication

Outcome / Impact: The system enhances visibility into network activity by linking it directly to system processes, enabling faster detection of malicious communication and improving incident response capabilities.

Security Platforms & Products

Projects and platforms focused on real-world cybersecurity operations, including incident response, threat detection, and large-scale forensic investigation workflows.

8. Cipher Triage – Rapid Forensic Triage & System Monitoring Platform

Overview: Worked with Cipher Triage, a rapid forensic triage and system monitoring platform designed to identify illicit content and analyze system activity on Windows environments. The platform enables investigators to quickly detect malware, suspicious user activity, and hidden system artifacts in both field and lab settings.

Approach / Implementation

- Detection of illicit and suspicious content (malware, user artifacts, system data)
- Real-time system monitoring including CPU, memory, disk, and network usage
- Visibility into running processes and active system behavior
- Monitoring of uptime and system stability
- Early detection of anomalies and performance issues

Key Features

- Rapid forensic triage and investigation
- Unified system and network visibility
- Real-time monitoring and anomaly detection
- Deep artifact analysis with modular approach
- User-friendly interface for both technical and non-technical users

Outcome / Impact: Cipher Triage enables security teams to quickly detect, analyze, and respond to cyber incidents by providing comprehensive system and network visibility. It significantly reduces investigation time and improves decision-making during critical incident response scenarios.

9. Sigma-Based Detection System – CDR/IPDR Analysis & Threat Intelligence Platform

Overview: Worked with a Sigma-based detection system designed for large-scale analysis of Call Detail Records (CDR) and Internet Protocol Detail Records (IPDR). The platform enables investigators and intelligence teams to process, correlate, and analyze communication and network data to uncover hidden relationships and suspicious activity.

Approach / Implementation

• Implemented Sigma-based rule-driven detection mechanisms for identifying suspicious patterns
• Designed workflows for ingesting large-scale CDR and IPDR datasets from multiple providers
• Developed correlation logic linking calls, messages, and IP sessions
• Enabled filtering and querying for rapid isolation of relevant events
• Integrated structured workflows for consistent investigation processes

Data Analysis & Visualization

• Interactive timelines for event reconstruction and sequence analysis
• Dynamic charts and network graphs for identifying hidden relationships
• Visualization of communication patterns and behavioral trends
• Real-time filtering and exploration of large datasets

Technologies / Concepts: Python, Pyside6**,** Sigma Rules, Data Correlation Techniques, Large-Scale Data Processing, Visualization Systems, Threat Intelligence Analysis

Key Features

• Large-scale CDR/IPDR data processing
• Rule-based detection using Sigma
• Cross-data correlation for intelligence insights
• Advanced filtering and querying capabilities
• Visual dashboards for investigation workflows

Outcome / Impact: The system enables rapid and scalable analysis of communication and network data, helping investigators uncover hidden connections, reconstruct timelines, and generate actionable intelligence. It significantly improves the efficiency and accuracy of data-driven investigations.

10. Cipher Sentinel Imager

Overview: Worked with Cipher Sentinel Imager, a comprehensive forensic acquisition solution designed for secure disk imaging, memory acquisition, and evidence verification. The tool supports both local and enterprise environments, enabling reliable and high-integrity digital evidence collection.

Approach / Implementation

• Supported disk imaging in multiple formats including E01, RAW, and DD
• Enabled RAM dump acquisition for volatile memory analysis
• Implemented drive cloning for forensic duplication of storage media
• Designed workflows ensuring reliable and efficient evidence collection

Evidence Integrity & Verification

• Implemented image verification using cryptographic hash algorithms (MD5, SHA1, SHA256)

• Validated acquired images against original data to ensure authenticity
• Displayed verification metrics including progress, speed, and time estimation
• Ensured integrity of forensic evidence for legal and investigative use
• Strengthened chain of custody through reliable validation mechanisms

Technologies / Concepts: Digital Forensics, Disk Imaging, Cryptographic Hashing, Memory Acquisition, Python, Qt5, Osfmount

Key Features
• Multi-format disk imaging (E01, DD, RAW)
• RAM dump acquisition
• Cryptographic image verification
• Network-based forensic acquisition
• Secure disk wiping with industry standards
• High-speed and reliable data acquisition

Outcome / Impact: The tool ensures accurate and secure forensic acquisition while maintaining strict evidence integrity. It enables investigators to confidently collect, verify, and manage digital evidence across various environments, supporting both investigative and legal requirements.

Systems & Software Engineering

Projects focused on building high-performance systems and software applications, demonstrating strong experience in real-time processing, system-level programming, and production-grade software development.

11. Multi Viewer – Real-Time Video Monitoring System

Overview: Developed a real-time video monitoring system capable of displaying and managing multiple live video streams from various sources on a single interface. The system supports multiple streaming protocols and is designed for high-performance video processing and monitoring.

Approach / Implementation
• Integrated multiple video input sources including UDP, TCP,HLS RTMP, and SDI
• Used FFmpeg for video decoding and stream processing
• Utilized TSDuck for transport stream analysis
• Developed GUI using MFC and Win32 API for Windows systems
• Implemented synchronization and performance optimization for smooth playback
• Designed robust error handling for uninterrupted monitoring

Technologies Used: C++, Python, FFmpeg, TSDuck, MFC, Win32 API, Streaming Protocols (UDP, RTMP, SDI).

Key Features
• Multi-stream video monitoring in real time

- Support for multiple streaming protocols
- High-performance video decoding
- Stream synchronization and stability
- Scalable and efficient system design

Outcome / Impact: The system enables efficient real-time monitoring of multiple video streams, improving operational visibility and reliability in broadcast and surveillance environments.

12. KOSTAC PLC Design Software

Overview: Worked on the development and enhancement of PLC design software used in industrial automation systems, focusing on validation, system integrity, and efficient design workflows.

Key Work

- Implemented validation logic for PLC configurations
- Ensured uniqueness and correctness of PLC identifiers
- Contributed to development, testing, and deployment lifecycle
- Improved system reliability and performance

Technologies: C++, MFC, Win32 API, Industrial Software Systems

Impact: Improved the reliability and integrity of PLC design processes, supporting industrial automation workflows.

13. Asrun Generator – Broadcast Scheduling Tool

Overview: Developed a Python-based application to generate broadcast scheduling (Asrun) data by interacting with databases and producing outputs in multiple formats for different channels.

Key Work

- Developed console-based application for Asrun generation
- Integrated database interaction for data retrieval
- Supported multiple output formats for broadcasting workflows
- Automated scheduling data generation

Technologies: Python, Database Systems, Data Processing

Impact: Streamlined the generation of broadcast scheduling data, improving efficiency and reducing manual effort.